

Too Good to be False: How Photorealism Promotes Susceptibility to Misinformation

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Abstract

Recent advances in generative AI enable the production of photorealistic images that blur the line between authentic and synthetic content, raising concerns about misleading users when misused. We examined how the realism of AI-generated images affects users' credibility judgments of misinformation and explored two key cognitive mechanisms: realism heuristic and synthetic heuristic. A between-subjects experiment ($N = 253$) revealed that highly realistic AI-generated images significantly increased the credibility of misinformation compared to less realistic images, and this effect was primarily driven by the realism heuristic, as photorealistic images were perceived as portraying reality. Additionally, photorealistic images tended to inhibit activation of the synthetic heuristic, which we newly proposed in this paper (i.e., the perception that images are artificially created or edited). Contrary to previous findings, our data show that regular users of generative AI and social media are more susceptible to misinformation. We propose identifying and labeling AI involvement in image creation and increasing literacy.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**.

Keywords

AI-Generated Misinformation, Photorealism, Realism Heuristic, Synthetic Heuristic, Credibility, Prior Experience

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1 Introduction

Recent advancements in generative AI produce realistic visuals that have increased the potential for synthetic content to mislead users [5]. While early generative AI models struggled to create convincing images, newer models, such as the latent diffusion model in text-to-image technologies, have dramatically improved the realism of synthetic visuals by creating photographic-quality images [2, 13]. Although these capabilities can benefit numerous applications, they also pose risks, as malicious actors can now generate realistic images that easily deceive others. This issue is particularly urgent because visuals are generally more persuasive than text-only modality [27], and research has shown that individuals are more likely to believe misinformation when paired with visual modality than text modality alone [27]. As generative AI becomes more sophisticated and accessible even to lay users, the consequences of realistic AI-generated images, including their potential to amplify the spread of misinformation, have become more evident [2, 34]. Scholars have also warned that this phenomenon could undermine public trust in media [2]. Despite its significant impact on our society, we still lack an understanding of how users perceive and evaluate misinformation paired with AI-generated images. In this study, we question whether users have developed a new cognitive mental shortcut that we call "synthetic heuristic," to suspect that the content is artificially manipulated or altered. We propose that this heuristic arises because of the surge of synthetic content online, with users being skeptical of online images rather than merely relying on realistic rendition of the visuals. Therefore, we investigate how image realism in AI-generated images shapes credibility judgments of misinformation and the underlying mechanisms (realism heuristic and synthetic heuristic) that may explain why individuals fall for misinformation and how susceptible they are to it.

2 Related Work

2.1 Image Realism in AI-Generated Media and Credibility Judgments

In the context of media, realism often refers to whether media content reflects the real world (factuality) [9], whether media representations could plausibly occur in reality (plausibility) [23], and

whether the visual elements of the media convincingly simulate reality with quality and vividness (perceived quality) [21], that is, how media representation aligns with the real world. Within this framework, images are a crucial modality that contributes to the realism of the media. Media realism has been widely studied, as individuals' perceptions of this realism can be a stronger predictor of behavioral outcomes than the attributes of the medium itself [22]. Building on this early work, we define image realism as the extent to which images are realistic and similar to reality, and feature graphical qualities that conform to real-world standards. Although, in the computer graphics domain, image realism specifically addresses the extent to which an image appears photo-realistic, we argue here that image realism is a broader concept than photorealism, encompassing a wide range of visuals including photography, CGI, and paintings [7].

The image realism produced by early text-to-image AI models lacked the sophistication to fool users into thinking that they portrayed reality. This was partly due to technical limitations and a nascent understanding of how to structure the models to create realistic images, including factors like hyperparameters, selection criteria, and effective text prompts [10]. Therefore, early AI-generated images were often evaluated as cartoonish and artistic paintings [10], making it easier for lay users to notice their departure from reality and surmise that they were artificially generated. However, recent advancements in generative AI have improved the realism of these images [10]. Previous studies have employed several metrics for evaluation, such as the Fréchet Inception Distance (FID) which measures the distributional similarity between synthetic and real images, and the Structural Similarity Index (SSIM), which offers an analysis of perceptual differences in luminance, contrast, and structure of image [33]. Using these metrics, the studies have revealed that systems such as DALL-E2 and Imagen consistently produce significantly more realistic images compared to alternatives such as Grok and Stable Diffusion [10, 13, 33]. These findings were also aligned with subjective human assessments of image quality and realism, such that users were more likely to view the latest generative AI images as more realistic compared to earlier versions [13]. Recent studies have also shown that users are often poor at detecting image manipulation and do not question the authenticity of forged images [16]. For example, users were less likely to question the authenticity of faces created using generative adversarial networks (GANs) than images of real faces [18], and were unable to distinguish synthetic facial images generated by GANs from real facial images [24]. These findings suggest that misinformation paired with realistic AI-generated images is likely to make users believe misinformation compared to one with less realistic images. Thus, we propose:

H1. Misinformation accompanied by highly realistic AI-generated images will seem more credible compared to misinformation with less realistic AI-generated images.

2.2 Realism Heuristic: Reflects Real World

When evaluating media content, users tend to rely on cognitive heuristics or mental shortcuts due to limited cognitive resources, especially with the proliferation of available information online. As cognitive misers [8], they process visual modalities, such as videos and images, more heuristically than text or audio, leading

to greater trust and sharing of visual content [27]. This trust is driven by the realism heuristic, which operates on the principle of "seeing is believing," leading users to believe that information with visuals is credible without much scrutiny [25]. For example, Sundar et al. showed that fake news shared on WhatsApp with a video modality triggered a greater realism heuristic than fake news with text or audio modalities, which made users superficially process the information. This then made users believe misinformation and want to share it with others [27].

While prior research has explored how the realism heuristic varies across modalities, such as text, images, audio, and virtual reality, differences can also occur within the same modality. For example, the realism of an image can be influenced by factors such as illumination, color, and saturation ([7]). Specifically, Fan et al. [7] found that computer-generated images were rated as less realistic than photographs. Similarly, earlier generative AI models, with their cartoon-like or artistic outputs, likely evoked less realism heuristic compared to advanced models that produce photo-realistic visuals with higher fidelity and detail. That is, the strong activation of the realism heuristic (that the portrayal of the content resembles the real world) evoked by highly realistic images can increase the credibility of the (mis)information. Thus, we propose:

H2 Misinformation accompanied by highly realistic AI-generated images will seem more credible compared to misinformation with less realistic AI-generated images because of the strong activation of the realism heuristic.

2.3 Synthetic Heuristic: Created Artificially

While realistic images can trigger the realism heuristic that makes users believe that the portrayal of the media content conforms to the real world, the proliferation of known deepfakes and other photo-shopped and/or AI-generated images online, as well as the publicity surrounding them, has likely led users to develop a new (complementary) cognitive shortcut, which we call: *synthetic heuristic*. We propose that this heuristic prompts users to question whether the content, despite appearing realistic, is synthetic, in that it is doctored, manufactured, or artificially created, rather than natural, human-made, or original. This heuristic speaks to users' skepticism about the nature of the content creation process, without regard for the realism of content portrayal. The activation of synthetic heuristic has significant implications for how users process AI-generated misinformation. If users fail to activate the synthetic heuristic when encountering misinformation paired with realistic AI-generated images, they may be less skeptical and more susceptible to believing misinformation. In sum, the synthetic heuristic can serve as an indicator of how vigilant users are with digital content.

Despite providing photorealistic-quality images, generative AI still produces visual imperfections or glitches [3]. These glitches range from obvious flaws, such as hands with four or six fingers, to more subtle inconsistencies, mismatched elements, lighting errors, or minor artistic distortions [10, 13]. Such imperfections and inconsistencies in images could serve as cognitive cues, prompting users to question whether the images are synthetic and artificially created, produced, or manipulated. However, we argue here that even human-created images as well as images without any imperfections can activate the synthetic heuristic. For example, when a human takes a photograph of a person's face and touches it up with Photoshop to remove freckles and wrinkles, the image may look

realistic. Yet, the uncanniness of its perfection could evoke suspicion about its synthetic nature. That is, triggers for the synthetic heuristic are not confined to AI-generated content but extend to any visual that seems too perfect to be true or exhibits obvious or even subtle visual imperfections or glitches.

In sum, the realism heuristic refers to the extent to which media content resembles the real world, leading users to heuristically trust content that looks real. The synthetic heuristic is about the extent to which users question the authenticity of content based on its origin or creation process, particularly their ability to detect its synthetic generation. This then opens up the possibility that the realism heuristic and synthetic heuristic do not necessarily always operate in a trade-off relationship (i.e., when the realism heuristic is strongly activated, then the synthetic heuristic is not activated or weakly activated, or vice versa). That is, both heuristics can operate simultaneously. While the realism heuristic involves events depicted in misinformation that seem real rather than fake, the synthetic heuristic involves suspicion that the content is artificially rather than naturally generated. The realism heuristic is about perceptions of the content itself, while the synthetic heuristic is related to perceptions of the way the content was created. Therefore, it is possible that users might perceive an image as highly realistic as it resembles the real-world (activating the realism heuristic) while simultaneously suspecting it may be synthesized in some ways during the creation process (activating the synthetic heuristic). For example, a photorealistic image of Donald Trump with four fingers illustrates a case where both heuristics could be strongly activated. However, this is not to say that these two heuristics are conceptually orthogonal; rather, they may exist on a continuum of reality and share some degree of overlap. Given these competing possibilities and the lack of empirical evidence, we pose research questions instead of proposing hypotheses:

RQ1. Can image realism trigger the synthetic heuristic? If so, how does the synthetic heuristic affect the credibility of AI-generated misinformation?

RQ2. Do realism and synthetic heuristics operate in parallel, or does one dominate heuristic processing of AI-generated misinformation?

3 Method

3.1 Study Design, Participants, Procedure, and Stimuli

We employed a 2 (image realism: low vs. high) X 2 (misinformation topics: politics vs. disaster) between-subjects design. Different misinformation topics were employed for stimulus sampling purposes. Participants ($N = 253$) were recruited from an online crowdsourcing platform, Cloud Connect, with US census-matched demographics (gender, age, ethnicity, and race). The sample consisted of 50.2% females ($n = 127$), 49.0% males ($n = 124$), and 0.8% prefer not to say ($n = 2$). The average age was 45.1 years ($SD = 0.94$, ranging from 18 to 79). Monthly household income ranged from “\$0 to \$14,999” (1) to “\$200,000 or more” (7), with a $Mdn = 3.00$; \$50,000 to \$74,999, $SD = 1.63$. This exempt study was approved by an institutional review board before data collection.

After consenting to participate in the study, participants first responded to questions asking about their use of generative AI



Figure 1: Politics - High (Left) and Low (Right) Image Realism

(covariate). Participants were then randomly assigned to one of four conditions, where they viewed a screen-captured post, which was described as having been widely circulated on the X platform. The X post contained a news headline, a brief summary, and a relevant AI-generated image (see Figure 1 and Appendix B). Participants were required to spend at least 15 seconds on the post before proceeding to the questionnaire, which, in randomized order, contained the outcome variables (see measures below). To eliminate potential confounds, the blue check mark (which can indicate account authenticity) and engagement metrics (e.g., number of likes, views, and shares) on the X posts were removed. The stimuli had identical formatting (e.g., font size, news source, and color) across all conditions.

For the low image realism conditions, images that scored low on the perceived image realism measure from the pretests were used, while for the high image realism conditions, ones that scored high were used. Details on the creation and selection of stimuli images and misinformation topics can be found in Appendices A through C.

3.2 Measures

Since the measures concern the nature of either an image or the portrayal of events in misinformation, we highlighted the target of the statements for each variable in the questionnaire to help participants' understanding.

3.2.1 Perceived image realism. As a manipulation check, we measured participants' perceptions of the realism of the AI-generated images on a 7-point scale (1 = not at all, 7 = extremely). The items included: “The image in the X post reflects real-world appearances” and “The visual quality of the image in the X post is high” ($M = 4.29$, $SD = 1.72$, Cronbach's $\alpha = .91$).

3.2.2 Message Credibility. Three items from Appelman and Sundar (2016) were used to measure the credibility of the X post (misinformation) on a 7-point Likert scale (1 = not at all, 7 = extremely). The items were: The news on X post seemed... “believable,” “authentic,”

and “inaccurate (reversed)” ($M = 4.04$, $SD = 1.70$, Cronbach’s $\alpha = .86$).

3.2.3 Realism Heuristic. Five items were adapted from Sundar et al. (2021) to assess the perceived alignment of the misinformation to reality as a proxy for measuring the realism heuristic (1 = not at all, 7 = extremely). The items were: “The news presented seemed quite similar to what really happened,” “The news seemed legitimate,” “Events in the news were portrayed vividly,” “The news seemed realistic,” “The news seemed legitimate,” and “I can tell that the news is true and not made up” ($M = 3.31$, $SD = 1.38$, Cronbach’s $\alpha = .89$).

3.2.4 Synthetic Heuristic. To evaluate participants’ perception that the image was synthetic, a 7-point semantic differential scale with seven items that we developed was used. The items were: The image in the news seemed.. “manipulated/unedited,” “altered/unaltered (reversed),” “unnatural/natural looking,” “fabricated/genuine,” “synthesized/original,” “artificially created/authentically created,” and “AI-generated/human-made” ($M = 4.51$, $SD = 2.11$, Cronbach’s $\alpha = .98$). The variable was recoded so that higher values indicated higher perceptions of a synthetic nature. The development and validation of these measures of synthetic heuristic (especially in distinguishing the concept from realism heuristic) is reported in Appendix D.

3.2.5 Prior Experience with Generative AI. Prior experience with AI is a key factor affecting individuals’ evaluations of AI-related content. To control this effect, participants were asked how often they use generative AI (e.g., ChatGPT, Gemini, Copilot, Dall-E, etc.) on a 5-point Likert scale (1 = never, 5 = almost every day). ($M = 2.99$, $SD = 1.30$).

3.3 Analysis Plan

We conducted a set of mediation analyses using PROCESS Macro (Model 4; [11]) with 5,000 bootstrapped samples to generate 95% confidence intervals (CIs). High image realism was coded as 1, and low image realism as 0. Misinformation topics (1 = politics, 0 = disaster), along with prior experience with generative AI, were entered as covariates.

4 Results

4.1 Manipulation Check

The independent t-test results showed that participants in the high image realism condition reported higher perceived image realism ($M = 5.47$, $SD = 1.01$) than those in the low image realism condition ($M = 3.12$, $SD = 1.45$), $t(251) = -14.97$, Cohen’s $d = 1.25$, $p < .001$. Thus, the manipulation was deemed successful.

4.2 H1: Highly Realistic AI-Generated Images Increased Perceived Credibility of Misinformation

H1 predicted that highly realistic AI-generated images would lead users to fall for misinformation more than less realistic AI-generated images when presented with misinformation. The results showed that participants exposed to misinformation paired with highly realistic AI-generated images perceived the misinformation as more

credible and believable, compared to misinformation paired with less realistic images ($b = 1.46$, $p < .001$), supporting H1.

4.3 H2: Highly Realistic AI-Generated Images Activated a Strong Realism Heuristic

H2 posited that the effect of image realism on credibility would be mediated by the realism heuristic. As predicted, highly realistic AI-generated images elicited a strong activation of the realism heuristic compared to less realistic ones ($b = 1.04$, $p < .001$), which in turn, was positively associated with message credibility of misinformation ($b = 0.93$, $p < .001$). This indirect effect was significant (*indirect effect* = 0.97, *BootSE* = 0.14, 95% CI [0.70, 1.24]), supporting H2.

4.4 RQ1: Highly Realistic AI-generated Images Impeded the Activation of Synthetic Heuristic

Highly realistic AI-generated images evoked lower levels of synthetic heuristic, compared to less realistic ones ($b = -3.21$, $p < .001$), which in turn was negatively associated with the credibility of misinformation ($b = -0.13$, $p = .003$). This indirect effect was significant (*indirect effect* = 0.43, *BootSE* = 0.15, 95% CI [0.14, 0.74]). This indicates that highly realistic AI-generated images decreased the activation of synthetic heuristic, thereby making users more vulnerable to misinformation.

4.5 RQ2: Realism Heuristic as a More Dominant Driving Force than Synthetic Heuristic

RQ2 asked how realism and synthetic heuristics operate—whether it is in parallel or with one more dominant than the other. Although both the realism and synthetic heuristics positively mediated the relationship between image realism and the credibility of misinformation, the results showed that the realism heuristic (RH) emerged as a dominant mediator ($b = 0.97$ for RH, $b = 0.43$ for SH; *difference* [RH - SH] = 0.54, *BootSE* = 0.21, 95% CI [0.13, 0.95]). This suggests that highly realistic AI-generated images led participants to perceive misinformation as more accurate and credible primarily by evoking the realism heuristic that the news events portrayed aligned with reality. The full coefficients of indirect effects are presented in Figure 2.

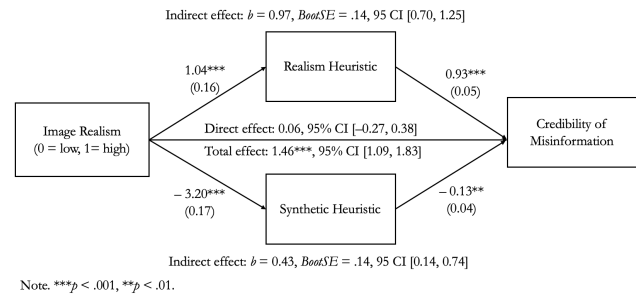


Figure 2: Indirect Effects via Realism and Synthetic Heuristics

5 Discussion

We investigated how the realism of AI-generated images shapes credibility judgments of misinformation and examined two underlying mechanisms: the realism heuristic and the synthetic heuristic. Below, we discuss the theoretical and practical implications of our findings as well as limitations and suggestions for future research.

5.1 Highly Realistic AI-Generated Images Inhibits the Activation of Synthetic Heuristic

First, the findings contribute to the literature on users' reliance on cognitive heuristics by demonstrating the effect of image realism on users' credibility judgments of misinformation. This effect was primarily due to the strong activation of the realism heuristic, making users rely heavily on visual cues to assess the credibility of the information, particularly when those cues align with their expectations of real-world appearances, supporting previous findings (e.g., [27]). Moreover, the findings showed that the synthetic heuristic was less likely to be activated when images were highly realistic, indicating that as generative AI advances its quality in creating images, users' ability to discern the synthetic nature of the media content has become weaker. Given that images in high realism conditions in this study exhibited some visual glitches, such as text gibberish on signs in the disaster topic, it appears that even imperfect AI-generated content may effectively deceive users as long as it appears realistic at a glance.

Although we found a more dominant role of the realism heuristic over the synthetic heuristic with a parallel mediation (Model 4), a serial mediation model (Model 6) in which both heuristics were tested as sequential mediators between image realism and credibility was also significant (path through RH $\rightarrow$ SH: $b = 0.30$, $BootSE = 0.10$, 95% CI [0.10, 0.53]; path through SH $\rightarrow$ RH: $b = -1.50$, $BootSE = 0.38$, 95% CI [-2.27, -0.77]). This indicates that either realism or synthetic heuristic can be triggered by the initial activation of the other heuristic, providing evidence that they may operate in a trade-off relationship. Together, the key finding is that when realistic AI-generated images trigger the realism heuristic, it may suppress the potential for triggering the synthetic heuristic. Therefore, more emphasis needs to be placed on identifying and labeling AI's involvement in the creation of images [4, 6, 32], to design interfaces that trigger the synthetic heuristic, which we discuss in detail in the following section.

5.2 Design Suggestions to Trigger Synthetic Heuristic

Advanced models that generate highly realistic images mimicking real-world appearances make users bypass the synthetic heuristic, which was easily activated by images generated by early models. Practitioners may develop interventions that can reactivate this synthetic heuristic, such as sharing the telltale signs of AI generation, or providing overlays over suspected AI-generated misinformation to highlight what exactly seems suspicious in an image. Such "labeling" of AI-generated content has proven effective in decreasing the likelihood of users engaging with misleading information [32]. As our data showed that detecting the source of the content

as AI was negatively associated with the credibility of misinformation, labeling the image source can prevent users from blindly trusting synthetic images. This becomes more important because images used in misinformation are not necessarily fake, such as AI-generated ones. It is important to note that real images from the past could also be repurposed to mislead people.

Beyond AI labeling, educators, media outlets, and policymakers may also consider media literacy interventions to improve users' AI literacy. This includes educating users about generative AI's abilities, such as how sophisticated and convincing images it can create, in safe spaces. One prominent example of this is *the New York Times's* "Test Yourself: Which Faces Were Made by A.I.?" quiz [28], which exposes readers to multiple AI-generated vs. real faces and asks them to guess the source of images. This practice can help users reduce overconfidence in their ability to discern AI-generated images, making them vigilant when evaluating synthetic images. Additionally, watermarking for AI-generated images with visible text like "created with AI" is a clear way to alert users. We invite stakeholders to develop standards for AI watermarks to ensure consistency across platforms.

5.3 More Experience with Generative AI, More Susceptibility to Misinformation?

Although it was not hypothesized, our data revealed that users who regularly use generative AI (i.e., greater prior experience with generative AI) were more susceptible to misinformation (i.e., they perceived misinformation credible) than those who do not ($b = 0.18$, $p = .024$). This finding contradicts previous research suggesting that AI literacy or familiarity can serve as an effective intervention against misleading content [14, 31]. One possible explanation is that frequent exposure to generative AI may normalize synthetic content, undermining users' ability to critically evaluate its authenticity, especially when the outputs align with real-world expectations. Habitual users may have developed cognitive overconfidence, believing that they can discern real from synthetic content, which inadvertently increases their susceptibility to misinformation.

Furthermore, our data showed that participants who regularly use X (formerly Twitter; measured with 1 = never, 5 = use it almost every day; $M = 2.89$, $SD = 1.65$) were also more vulnerable to misinformation ($b = 0.17$, $p = .005$). This is analogous to the finding in the multitasking literature [19] that heavy media multitaskers were less capable of filtering out irrelevant representations in memory. It is likely that heavy social media users, inundated with a vast amount of information, might experience similar cognitive strain, such as reducing their ability to critically assess social media posts [17]. Social media algorithms could further exacerbate this by prioritizing visually engaging, highly realistic content, regardless of its authenticity. These exploratory findings suggest how familiarity or regular use of generative AI and social media can inadvertently increase susceptibility to misinformation, emphasizing the need for tailored educational and platform-level interventions to address these challenges. One promising intervention would be employing literacy messages that forewarn users about potential misleading information or highlight the responsibility of being critical when consuming information online. For example, a previous study found that a simple news literacy message that tells citizens that it is their

job to critically consume and evaluate news effectively improved their news literacy [29].

5.4 Limitations and Suggestions for Future Research

While this study provides valuable insights, it has limitations. First, the study relied on pretested stimuli to create clear distinctions between low and high image realism. However, real-world misinformation exists on a continuum of realism, affected by additional factors such as the credibility of the source or the emotional appeal of the content. Future research should explore how these factors play a role in the proposed relationships. Second, as this was a one-shot study, it is hard to establish causal relationships between the mediators (heuristics) and the dependent variable (credibility), especially when all three variables were measured simultaneously. This design may have also prompted participants to engage in more deliberative reasoning rather than relying on heuristics. Given that shifting attention to accuracy can undermine the credibility of misleading information [20] and that news articles with images enhance users' selective engagement with the content [15], future studies may consider a more naturalistic design to examine how people make credibility judgments. However, because our study design focused on accuracy, yet participants still perceived misinformation as credible, our findings may underestimate the extent to which users believe misinformation in digital spaces. Third, as generative AI becomes more integrated into daily life, users may have developed more nuanced strategies for recognizing (or not recognizing) synthetic content, as seen in our exploratory findings. Moreover, users are not only cognitive misers but also rely on the satisficing principle [1], meaning they may consider synthetic images sufficiently credible for their purposes (e.g., synthetic images being used in truthful content). However, a negative association between the synthetic heuristic and the credibility of misinformation in our study suggests that when users detect synthetic cues, it can undermine the content's perceived credibility, potentially prompting them to engage in more critical thinking. Lastly, while our data suggest that regular users of generative AI may be more susceptible to AI-generated misinformation, our one-shot, small-sample study limits generalizability. Further research is needed to examine how prior AI experience shapes reliance on heuristics for making credibility judgments.

6 Conclusion

Our findings reveal that highly photo-realistic AI-generated images activate the realism heuristic while suppressing the synthetic heuristic, making users more susceptible to misinformation. As AI-generated content becomes increasingly convincing, users' ability to detect synthetic media weakens. Thus, we suggest potential interventions such as AI labeling (particularly the source of the image), media literacy programs, and standardized watermarking. Additionally, our findings suggest that frequent exposure to generative AI and social media may inadvertently increase susceptibility to misinformation, challenging the traditional view that familiarity or literacy fosters critical evaluation skills. Future research should

explore how users adapt to AI-generated content and identify effective strategies to trigger the synthetic heuristic that may alert users to potential misinformation online.

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7 Appendices

A Pretests for Stimuli Creation

We conducted a round of pretests to determine (1) misinformation topics, (2) low vs. high image realism of the stimuli images, and (3) source credibility of *Poynter*. First, we used Snopes.com (a fact-checking website) to identify misinformation topics flagged as fake or misleading content. We also employed ChatGPT to create or edit misinformation. A randomized order of 10 fake news headlines with brief summaries (three news topics as a between-subjects factor), each kept at approximately 10 words for the headline and 35–40 words for the summary, respectively, was presented. Pretest participants were asked to rate the credibility of the news, its newsworthiness, and sensationality. The final set of misinformation topics was selected based on the following criteria: (1) news stories evaluated as borderline in terms of message credibility (believable, authentic, and accurate) on a 7-point Likert scale from 1 = strongly disagree, 7 = strongly agree and (2) those that did not differ significantly in terms of sensationality. The newsworthiness measures (important, timely, relevant, interesting, serious, and informative; [26]) were unreliable (Cronbach's Alpha .40 to .60), and thus were not used as a criterion.

Second, we created images relevant to the misinformation topics chosen from the first pretest. We used a customized GPT model called *Photo Realistic Image GPT*, *Photo Realistic Image GPT - Pro*, *Midjourney*, *Stable Diffusion - Turbo model*. Pretest participants were asked to respond to gauge perceived image realism: “The image reflects real-world appearances,” “The image could plausibly exist in reality,” “The visual quality of the image is high,” “The image

looks authentic,” and “The image aligns with my expectations of a real-world scene,” on a scale from 1 = *not at all* to 7 = *extremely*. The final images were selected based on participants' ratings, ensuring a clear distinction between highly realistic and less realistic images as perceived by the participants.

Although the news source was constant across conditions, the source itself can play a significant role in evaluating credibility [12]. Therefore, the source *Poynter* was evaluated. According to AllSides' Media Bias Chart, *Poynter* is positioned at the center of the political spectrum, and it is known that many people outside the media industry or journalism studies may not be familiar with it. Participants responded to the statements “I know the news source, *Poynter*” ($M = 2.06$, $SD = 1.74$) and “I can trust the source, *Poynter*” ($M = 3.57$, $SD = 1.34$) using a scale from 1 = *not at all* to 7 = *extremely*.

B Disaster - Stimuli X Posts

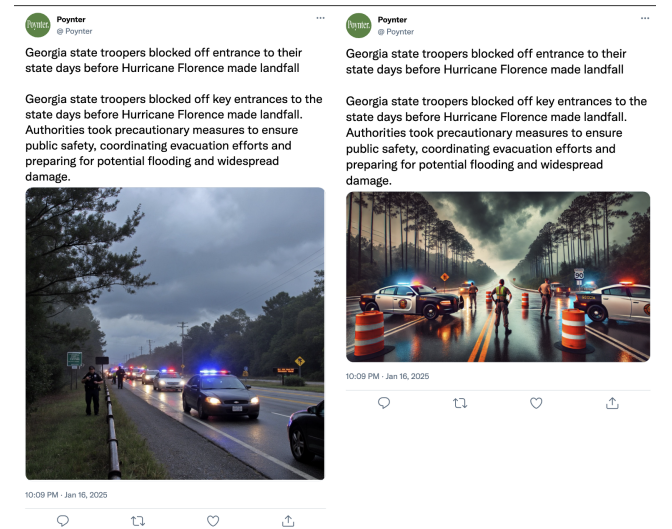


Figure 3: Disaster - High (Left) and Low (Right) Image Realism

C Prompts for Image Generation

To generate images, we used the news headline and a brief summary as a prompt. Additionally, we employed *Midjourney V6.1 - Photorealistic Image Prompts (Customized ChatGPT)* to create a detailed prompt for image generation.

C.1 Politics

C.1.1 News Headline + Brief Summary. "Police Raid and Burn a Standing Rock Protest Camp." Authorities raided and burned a protest camp at Standing Rock, dismantling resistance against a controversial pipeline project. The action has sparked outrage, with activists condemning the use of force, while officials cite safety concerns and legal orders."

C.1.2 Midjourney V6.1 - Photorealistic Image Prompt. "A dramatic wide-angle shot of a police raid on a Standing Rock protest camp, officers in riot gear moving through the smoky remains of burned

tents, protesters standing in defiance with raised fists, embers glowing in the air, a sense of tension and resistance, late afternoon golden light filtering through the smoke, shot with a Sony A7S III, 28mm wide-angle lens, high contrast with deep shadows and warm tones."

C.2 Disaster

C.2.1 News Headline + Brief Summary. "Georgia state troopers blocked off entrance to their state days before Hurricane Florence made landfall." Georgia state troopers blocked off key entrances to the state days before Hurricane Florence made landfall. Authorities took precautionary measures to ensure public safety, coordinating evacuation efforts and preparing for potential flooding and widespread damage.

C.2.2 Midjourney V6.1 - Photorealistic Image Prompt. "A bird's eye view shot of an emergency evacuation scene in Southern Utah, featuring residents hurriedly leaving their homes as rescue teams coordinate efforts amidst a backdrop of rugged desert terrain, dark storm clouds looming overhead, creating a dramatic and tense atmosphere, during late afternoon with soft golden light contrasting with the road."

D Development and Validation of Synthetic Heuristic

The authors developed the concept of the synthetic heuristic and its measures based on our theoretical definition (see Sections 2.3 and 3.2.4). Since it focuses on the authenticity of content origins and creation, we employed adjectives with opposing values. The last item, "AI-generated/human-made," was designed to capture perceptions of content authenticity by attributing its source as previous research found that users tend to undermine its credibility when discovered the source was AI [30]. However, we acknowledge that "human-made" may not always be perceived as the direct opposite of "AI-generated," because the content altered using tools like Photoshop could still trigger the synthetic heuristic despite being technically "human-made," as discussed.

To determine whether the synthetic heuristic is a distinctive concept from the realism heuristic, we performed a confirmatory factor analysis (CFA) using R - *lavaan* package. In our analysis, we tested two models: (1) a single-factor second-order model, in which both concepts loaded onto a single latent factor, and (2) a two-factor first-order model, in which the synthetic heuristic and the realism heuristic were treated as separate factors.

The results showed that the two-factor first-order model ($\chi^2[53] = 144.06, p < .001, CFI = .98, RMSEA = .08$ with 90%CI[.07, .10], SRMR = .03) provided a significantly better fit than the single-factor second-order model ($\chi^2[54] = 755.23, p < .001, CFI = .84, RMSEA = .23$ with 90%CI[.21, .24], SRMR = .14), $\chi^2(1) = 611.15, p < .001$. The CFA results provided evidence that the synthetic heuristic is conceptually distinct from the realism heuristic.